

Department of Computer Science and Engineering Assignment

Question

Course Code & Name : DSA0402 – FUNDAMENTALS OF DATA SCIENCE

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Assignment Title:	Bank Customer Subscription Prediction and Customer Segmentation System
Course Outcome(s) - CO:	1. Analyze the role of data science, facets of data, and the big data ecosystem for informed decision-making. 2. Apply data preprocessing and visualization techniques using Python libraries for effective data handling. 3. Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. 4. Develop machine learning models for classification, regression, and clustering, including performance evaluation and methodology comparison. 5. Design and implement end-to-end data-driven solutions by integrating preprocessing, statistical analysis, and machine learning for real-world applications
Bloom's Taxonomy Level	L3 - Apply; L4 - Analyse; L5 - Evaluate; L6 - Create
SDG Mapping (SDG 1-17)	SDG 8 - Decent Work and Economic Growth SDG 9 - Industry, Innovation and Infrastructure SDG 12 - Responsible Consumption and Production

Problem Statement :

A banking institution wants to understand its customers and improve the effectiveness of its marketing campaigns. The institution has historical customer information, demographic attributes, financial details, and previous marketing campaign outcomes. Develop a **Data Science and Machine Learning system** that predicts whether a customer is likely to subscribe to a bank term deposit and also groups customers with similar characteristics to support targeted marketing strategies.

The proposed system should demonstrate the complete **Data Science process**, including data retrieval, data preprocessing, exploratory data analysis, statistical analysis, supervised learning, model evaluation, and unsupervised learning.

Technical Requirements

- Select an appropriate **publicly available bank marketing/customer dataset** and clearly mention its source.

- Perform necessary **data preprocessing** using Python and Pandas, including:
 - Reading and inspecting the dataset.
 - Identifying and handling missing or inconsistent data.
 - Removing irrelevant or redundant attributes.
 - Converting categorical attributes into suitable numerical representations.
 - Filtering and selecting relevant records and attributes.
 - Sorting and grouping data wherever appropriate.
 - Preparing the final dataset for statistical and machine-learning analysis.
 - Perform suitable **Exploratory Data Analysis (EDA)** to understand important characteristics of the dataset, including:
 - Data summarization.
 - Distribution of important variables.
 - Identification and treatment of potential outliers.
 - Analysis of categorical and numerical attributes.
 - Comparison of customer characteristics between subscribers and non-subscribers.
- Calculate appropriate **descriptive statistical measures**, including:
 - Mean.
 - Variance.
 - Covariance.
 - Correlation.
 - Suitable measures of data distribution and variability.
 - Investigate relationships between important customer attributes using **correlation analysis** and provide suitable interpretations.
 - Where appropriate, formulate a simple statistical inference regarding an important customer characteristic and demonstrate the use of:
 - Point estimation, and/or
 - Confidence intervals.
 - Divide the dataset appropriately into **training and testing data** for supervised learning.
 - Implement at least **three supervised learning algorithms** , such as:
 - **k-Nearest Neighbours (kNN) Classifier**
 - **Decision Tree / CART**
 - **Logistic Regression**
 - Train the selected models to classify customers into:

Subscribed / Not Subscribed

- Select appropriate input attributes based on the characteristics of the dataset and justify their selection.

- Evaluate the implemented classification models using suitable **evaluation metrics**, such as:
 - Accuracy.
 - Precision.
 - Recall.
 - F1-score.
 - Confusion Matrix.
 - Compare the performance of the implemented **kNN, Decision Tree/CART, and Logistic Regression** models.
 - Identify the **best-performing model** based on suitable evaluation measures and provide a clear justification for the selection.
 - Apply **K-means clustering** to group customers based on relevant demographic, financial, and/or behavioural attributes.
 - Determine an appropriate number of clusters using a suitable method and justify the selected value of **K**.
 - Analyze and interpret the characteristics of each customer cluster, for example:
 - High-value customers.
 - Young/low-income customers.
 - Customers with strong campaign engagement.
 - Customers with low subscription probability.
 - Compare the identified customer segments with the predicted subscription behaviour obtained from the classification models. Identify customer segments with high potential for subscription and/or high-risk or low-response segments.
 - Provide meaningful visualizations using **Matplotlib** for:
 - Distribution of important numerical variables.
 - Categorical variable frequencies.
 - Outlier analysis.
 - Correlation analysis.
 - Classification model performance.
 - Confusion matrices.
 - K-means customer clusters.
 - Use the final results to provide **data-driven marketing recommendations** for different customer segments.
 - The complete implementation should be executable **from beginning to end without errors**, using appropriate Python libraries such as **NumPy, Pandas, SciPy, Scikit-learn, and Matplotlib**.

Expected Outcome

The completed system should demonstrate a complete **Data Science workflow**:

Data Retrieval → Data Cleaning → Data Transformation → Data Exploration → Descriptive Statistics → Statistical Inference → Model Building → Prediction → Model Evaluation → Model Selection → K-means Clustering → Customer Segmentation → Interpretation → Recommendations

Students should be able to demonstrate how statistical analysis and machine-learning techniques can be applied to a real-world banking dataset.

The selected concepts directly reflect course emphasis on **data preprocessing, exploratory data analysis, estimation, statistical inference, supervised learning, clustering, and evaluation metrics**.

Deliverables

To receive full credit, each submission must include the following deliverables:

- **Pseudocode / workflow** of the proposed Data Science and Machine Learning solution.
- **Dataset description and source**, including:
 - Dataset name.
 - Number of records.
 - Number and description of attributes.
 - Target variable.
 - Source of the dataset.
- **Data preprocessing report**, including:
 - Missing-value handling.
 - Data cleaning.
 - Attribute selection.
 - Encoding of categorical variables.
 - Data transformation.
 - Outlier treatment, wherever required.
- **Exploratory Data Analysis**, including appropriate tables, statistics, and graphs.
- **Descriptive statistical analysis**, including appropriate measures such as:
 - Mean.
 - Variance.
 - Covariance.
 - Correlation.
- **Statistical inference**, including at least one appropriate point estimate or confidence interval, where applicable.
- **Python implementation/code** using appropriate libraries.
- **Screenshots of important outputs and visualizations.**
- **Implementation of three supervised learning algorithms**:
 - kNN.
 - Decision Tree/CART.

- Logistic Regression.
- **Performance comparison** of the implemented models using suitable evaluation metrics.
- **Confusion matrices** and interpretation of classification results.
- **K-means clustering implementation**, including justification for the selected number of clusters.
- **Visualization and interpretation of customer clusters.**
- **Final observations and conclusion.**
- **Business/marketing recommendations** based on the identified customer groups and prediction results.
- **One-page individual contribution write-up** from every team member.

Assessment Rubrics (Total: 100 Marks)

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Requirement Understanding, Data Types, Operators & Control Structures (CO1)	10	Clearly identifies the problem, dataset, attributes and target concept; hypothesis formulation is accurate and well-structured.	Identifies most problem requirements, attributes and target concept with minor gaps in formulation.	Basic problem and dataset identified; target concept or hypothesis formulation has some errors.	Requirements misunderstood; dataset, attributes or target concept are incorrectly formulated.

Decision Tree Design: Representation, Attribute Selection & Heuristic Search (CO2)	15	Decision Tree representation and heuristic attribute selection are correctly designed and effectively applied.	Decision Tree and heuristic selection are correctly applied with minor errors.	Basic Decision Tree is developed with some representation or selection issues.	Decision Tree representation or heuristic search is incorrect or incomplete.
Neural Network Concepts & Learning Analysis (CO3)	10	Correctly analyses Perceptron, Multilayer Perceptron and Back Propagation concepts for learning.	Analyses neural-network concepts with minor gaps or errors.	Demonstrates basic understanding of neural-network learning.	Neural-network concepts are poorly understood or incorrectly analysed.
Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Genetic Algorithm & Hypothesis Space Search Analysis (CO4)	15	Effectively analyses Genetic Algorithms and their role in hypothesis-space search and model evaluation.	Correctly explains GA-based search with minor gaps.	Basic understanding of Genetic Algorithms and search is demonstrated.	Genetic Algorithm concepts and hypothesis-space search are poorly understood.
Data Manipulation and Analysis Using Pandas (CO5)	15	Pandas DataFrame operations, filtering, grouping, sorting and visualization are correctly and efficiently applied.	Pandas operations are correctly applied with minor errors or gaps.	Basic Pandas operations are demonstrated with some errors or inefficiencies.	Pandas operations are missing, incorrect or poorly implemented.

Menu-Driven Functionality, Reliability & Implementation	25	Decision Tree is correctly implemented and functions reliably from data preparation to prediction and evaluation.	Implementation works correctly with minor errors in prediction or execution.	Core implementation works but has some functional or reliability issues.	Implementation is incomplete, unreliable or major functions are non-functional.
Reflection: Design Decisions, SDG Relevance & Learning Outcomes	10	Thoughtful justification of design choices, clear connection to SDG 7/8/12 relevance, and honest, specific account of challenges faced and learning gained.	General justification of design choices; SDG 7/8/12 relevance and challenges are mentioned but lack depth.	Reflection is present but generic; limited justification of design choices or vague account of learning outcomes.	No genuine reflection submitted, or content is generic with no clear design, SDG 7/8/12 or learning connection.

Abstract

This project develops an end-to-end data science solution for a banking marketing problem. The system uses historical customer and campaign information to estimate whether a customer is likely to subscribe to a term deposit and to identify groups of customers with similar characteristics. The workflow follows the assignment requirements: data retrieval, preprocessing, exploratory analysis, descriptive statistics, statistical inference, supervised classification, evaluation, K-means clustering, interpretation, and marketing recommendations. Three required classifiers are implemented: k-Nearest Neighbours, Decision Tree, and Logistic Regression. The final model is selected using more than accuracy alone because the target is imbalanced. K-means is then applied to demographic, financial, and behavioural variables to produce actionable customer segments.

1. Introduction

The uploaded DSA0402 assignment asks for a practical data science and machine learning system based on a public banking dataset. The business objective is to improve marketing effectiveness by predicting term-deposit subscription and grouping customers into meaningful segments. The solution therefore treats the problem as a binary classification task combined with an unsupervised clustering task.

1.1 Problem Statement

A banking institution has historical demographic, financial, and campaign information. The proposed system predicts whether a customer will subscribe to a term deposit and groups customers with similar characteristics so that marketing actions can be targeted more efficiently.

1.2 Objectives

- Retrieve and inspect a suitable public bank marketing dataset.
- Clean, transform, filter, group, and encode the data using Pandas and NumPy.
- Perform EDA, outlier analysis, descriptive statistics, covariance, and correlation analysis.
- Estimate the overall subscription proportion and construct a 95% confidence interval.
- Train kNN, Decision Tree/CART, and Logistic Regression classifiers.
- Evaluate the models with accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices.
- Select the best model using a balanced interpretation of predictive performance.
- Use K-means and silhouette analysis to determine an appropriate number of customer segments.
- Compare segment subscription rates with the classification predictions.
- Convert the analytical findings into practical marketing recommendations.

2. Assignment Requirement Alignment

The implementation was designed directly from the uploaded five-page assignment. Pages 1–3 specify the complete data science workflow, the three classifiers, K-means, statistical measures, inference, visualizations, and business recommendations. Page 4 lists the required implementation outputs and individual contribution write-up. Page 5 additionally evaluates menu-driven functionality, Pandas operations, design decisions, SDG relevance, and reflection.

Assignment requirement	Implementation response
Dataset source and description	UCI Bank Marketing; bank-full.csv; 45,211 records; 16 predictors + target
Preprocessing	Duplicate removal, missing-value validation, type conversion, target encoding, categorical encoding
EDA	Distributions, frequencies, subscriber comparisons, outliers, correlation matrix
Descriptive statistics	Mean, variance, covariance, correlation, distribution measures
Statistical inference	Point estimate and 95% confidence interval for subscription proportion
Supervised learning	kNN, Decision Tree/CART, Logistic Regression
Evaluation	Accuracy, precision, recall, F1, ROC-AUC, confusion matrices
Clustering	K-means with K selected by silhouette score
Visualization	Matplotlib figures saved in outputs/figures
Pandas operations	Filtering, sorting, grouping, aggregation and export
Menu-driven functionality	Interactive console menu included in the Python implementation
Recommendations	Segment-specific marketing actions based on subscription behaviour
Individual contribution	One-page contribution template included at the end of this report

3. Dataset Description

The selected dataset is the UCI Machine Learning Repository Bank Marketing dataset. The UCI documentation describes the data as direct marketing campaign information from a Portuguese banking institution and states that the classification objective is to predict whether a client subscribes to a term deposit. The selected bank-full.csv version contains 45,211 records and 17 columns: 16 input attributes and the binary output variable y.

Source: UCI Machine Learning Repository, Bank Marketing, dataset 222. Citation: Moro, S., Rita, P., & Cortez, P. (2014), Bank Marketing Dataset, DOI 10.24432/C5K306.

3.1 Attributes

Attribute	Type	Description
age	Numeric	Customer age
job	Categorical	Occupation
marital	Categorical	Marital status
education	Categorical	Education level
default	Binary	Credit in default
balance	Numeric	Average yearly account balance
housing	Binary	Housing loan
loan	Binary	Personal loan
contact	Categorical	Communication method
day	Numeric	Day of last contact
month	Categorical	Month of last contact
duration	Numeric	Duration of last contact
campaign	Numeric	Number of contacts during current campaign
pdays	Numeric	Days since previous campaign contact
previous	Numeric	Number of previous campaign contacts
poutcome	Categorical	Previous campaign outcome
y	Binary target	Term-deposit subscription: yes/no

4. Proposed System Workflow

The implementation follows the sequence requested in the assignment. The model development stage is separated from clustering so that the supervised prediction task and customer segmentation task can be interpreted independently and then compared.

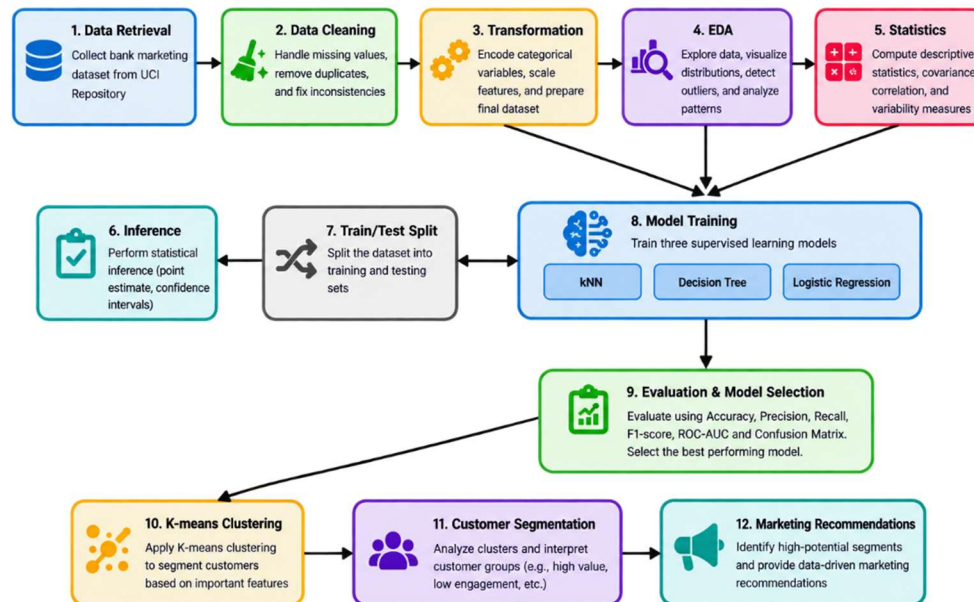


Figure 4.1: End-to-end workflow of the proposed bank customer subscription and segmentation system.

4.1 Pseudocode

1. START
2. Load the UCI bank-full.csv dataset.
3. Inspect shape, columns, data types, duplicates and missing values.
4. Clean strings and data types; remove duplicate records.
5. Treat unknown category values as explicit information rather than silently deleting them.
6. Encode the target: yes = 1 and no = 0.
7. Create descriptive statistics, covariance and correlation outputs.
8. Perform EDA and outlier analysis using Matplotlib.
9. Estimate the subscription proportion and calculate its 95% confidence interval.
10. Remove target and leakage-prone duration from the predictive feature set.
11. Split data into stratified training and testing sets.

12. Build kNN, Decision Tree and Logistic Regression pipelines with categorical encoding and scaling.
13. Evaluate each classifier and generate confusion matrices.
14. Select the strongest model using F1, recall and ROC-AUC together with accuracy.
15. Prepare demographic, financial and behavioural variables for K-means.
16. Test K values from 2 to 8 using silhouette scores.
17. Fit K-means with the selected K and profile each customer cluster.
18. Compare cluster subscription rates with predicted subscription behaviour.
19. Produce segment-specific marketing recommendations.
20. END

5. Data Preprocessing

The raw file is semicolon-separated. The program first standardizes column names and removes unwanted quote characters and surrounding spaces. Duplicate rows are identified and removed. Missing-value counts are measured before and after cleaning. The UCI page states that the dataset has no formal missing values; therefore, the value 'unknown' is kept as a legitimate category rather than being treated as a missing value by default.

5.1 Attribute Selection

The target y is separated from the predictor set. The variable duration is intentionally excluded from the pre-campaign prediction model. It records the duration of the current contact and is available only after the call occurs. Including it in a model intended for targeting before or at the beginning of a call can therefore introduce information leakage. It remains in EDA and descriptive statistics because it is useful for understanding the historical campaign.

5.2 Encoding and Transformation

Binary target: $y = \text{yes}$ becomes 1 and $y = \text{no}$ becomes 0.

Categorical predictors are one-hot encoded with unknown categories ignored at prediction time.

Numerical predictors are standardized inside the modelling pipeline.

The preprocessing transformer is fitted only on training data to avoid data leakage.

The target distribution is checked because subscription is the minority class.

5.3 Outlier Treatment

Potential outliers are visualized using boxplots. The implementation does not automatically delete extreme balances, campaign counts, or contact durations merely because they are unusual. For banking data, an extreme value can represent a real customer rather than an error. The report therefore distinguishes statistical unusualness from data invalidity.

6. Exploratory Data Analysis

EDA is used to understand the distribution of numerical variables, the frequency of important categories, and differences between customers who subscribed and those who did not. The program creates separate Matplotlib figures for age, balance, duration, campaign, job, education, housing, previous campaign outcome, subscriber comparisons, outlier analysis, and the correlation matrix.

6.1 EDA Figures

After execution, the generated PNG files in outputs/figures should be inserted below the relevant subsections. The filenames are intentionally descriptive so that screenshots can be matched to the report.

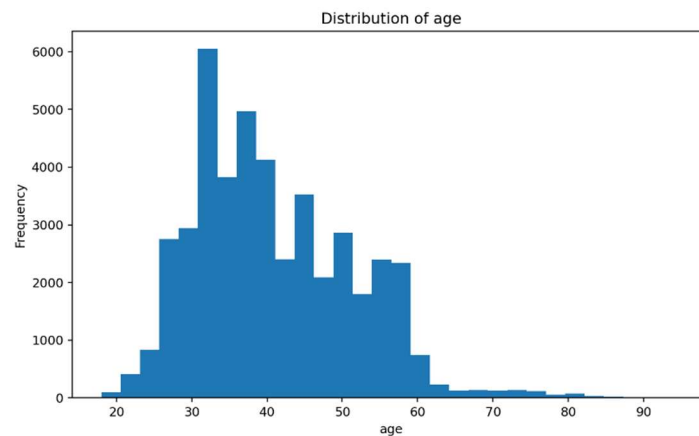


Figure placeholder: Age distribution.

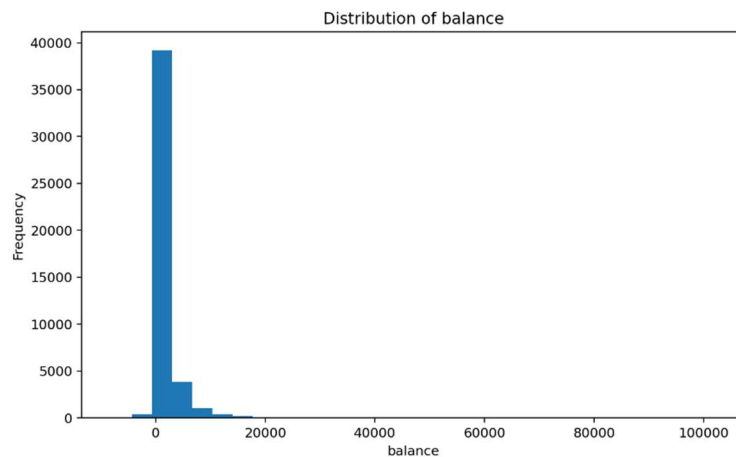


Figure placeholder: Balance distribution.

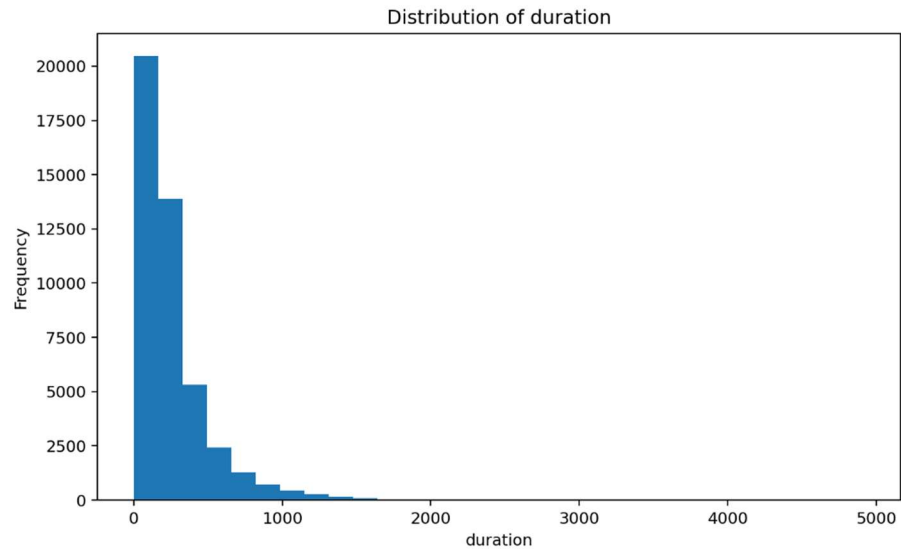


Figure placeholder: Current-contact duration distribution.

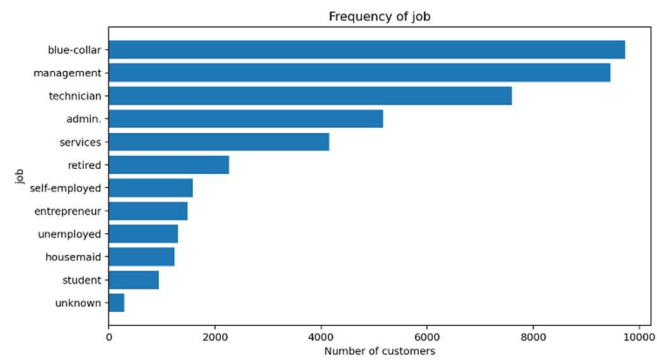


Figure placeholder: Job category frequencies.

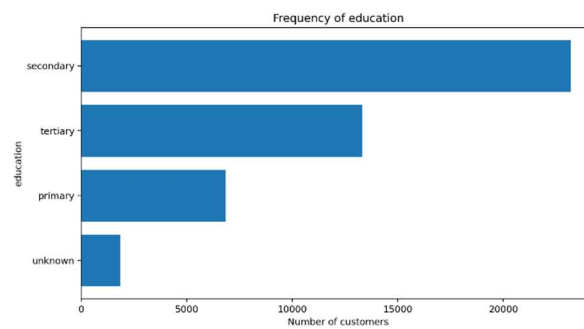


Figure placeholder: Education category frequencies.

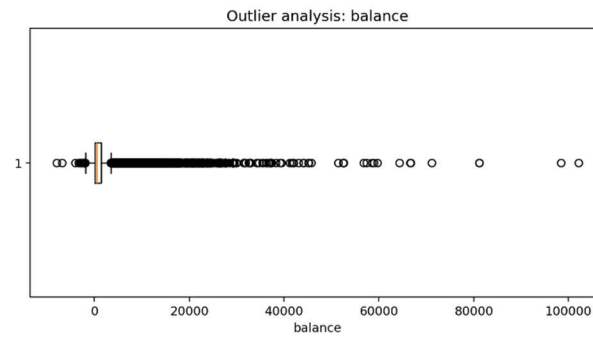


Figure placeholder: Balance outlier analysis.

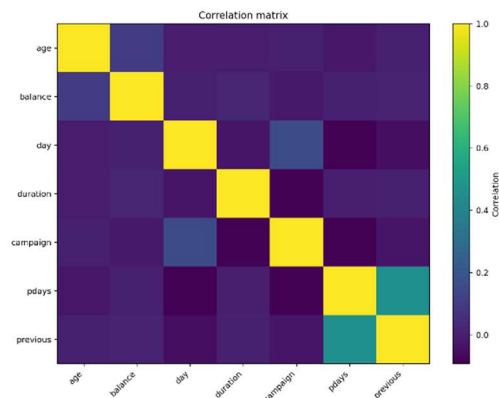


Figure placeholder: Correlation matrix.

7. Descriptive Statistical Analysis

The implementation calculates count, mean, standard deviation, minimum, quartiles, maximum, and variance for the main numerical variables. Covariance is calculated as a measure of joint variation, while Pearson correlation is used to examine linear relationships among numerical attributes.

The exact numerical values should be taken from `outputs/descriptive_statistics.csv`, `outputs/covariance_matrix.csv`, and `outputs/correlation_matrix.csv` after executing the program.

Measure	Variables / output	Final value																																																																																
Mean	Age, balance, day, duration, campaign, pdays, previous	<table><tr><th></th><th>count</th><th>mean</th></tr><tr><td>age</td><td>45211</td><td>40.9362102</td></tr><tr><td>balance</td><td>45211</td><td>1362.27205</td></tr><tr><td>day</td><td>45211</td><td>15.8064187</td></tr><tr><td>duration</td><td>45211</td><td>258.163079</td></tr><tr><td>campaign</td><td>45211</td><td>2.76384065</td></tr><tr><td>pdays</td><td>45211</td><td>40.1978279</td></tr><tr><td>previous</td><td>45211</td><td>0.58032337</td></tr></table>		count	mean	age	45211	40.9362102	balance	45211	1362.27205	day	45211	15.8064187	duration	45211	258.163079	campaign	45211	2.76384065	pdays	45211	40.1978279	previous	45211	0.58032337																																																								
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Distribution / variability	Quartiles, standard deviation, range	<table><tr><th></th><th>count</th><th>mean</th><th>std</th><th>min</th><th>25%</th><th>50%</th><th>75%</th><th>max</th><th>variance</th></tr><tr><td>age</td><td>45211</td><td>40.9362102</td><td>10.6187620</td><td>18</td><td>33</td><td>39</td><td>48</td><td>95</td><td>112.7581072828605</td></tr><tr><td>balance</td><td>45211</td><td>1362.27205</td><td>3044.76582</td><td>-8019</td><td>72</td><td>448</td><td>1428</td><td>102127</td><td>9270598.954472251</td></tr><tr><td>day</td><td>45211</td><td>15.8064187</td><td>8.32247615</td><td>1</td><td>8</td><td>16</td><td>21</td><td>31</td><td>69.2636093179959</td></tr><tr><td>duration</td><td>45211</td><td>258.163079</td><td>257.527812</td><td>0</td><td>103</td><td>180</td><td>319</td><td>4918</td><td>66320.57409008562</td></tr><tr><td>campaign</td><td>45211</td><td>2.76384065</td><td>3.09802088</td><td>1</td><td>1</td><td>2</td><td>3</td><td>63</td><td>9.597733393233844</td></tr><tr><td>pdays</td><td>45211</td><td>40.1978279</td><td>100.128745</td><td>-1</td><td>-1</td><td>-1</td><td>-1</td><td>871</td><td>10025.765773649764</td></tr><tr><td>previous</td><td>45211</td><td>0.58032337</td><td>2.30344104</td><td>0</td><td>0</td><td>0</td><td>0</td><td>275</td><td>5.305840647473837</td></tr></table>		count	mean	std	min	25%	50%	75%	max	variance	age	45211	40.9362102	10.6187620	18	33	39	48	95	112.7581072828605	balance	45211	1362.27205	3044.76582	-8019	72	448	1428	102127	9270598.954472251	day	45211	15.8064187	8.32247615	1	8	16	21	31	69.2636093179959	duration	45211	258.163079	257.527812	0	103	180	319	4918	66320.57409008562	campaign	45211	2.76384065	3.09802088	1	1	2	3	63	9.597733393233844	pdays	45211	40.1978279	100.128745	-1	-1	-1	-1	871	10025.765773649764	previous	45211	0.58032337	2.30344104	0	0	0	0	275	5.305840647473837
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8. Statistical Inference

A point estimate is obtained from the sample proportion of customers who subscribed. A 95% confidence interval is then calculated using the normal approximation to the binomial proportion. The interval provides a range of plausible values for the subscription rate in the population represented by the sampled records.

Statistic	Result
Sample size	45,211
Number subscribed	5,289
Estimated subscription proportion	0.116985
95% confidence interval	0.114022 – 0.119947

9. Supervised Learning Models

The assignment requires at least three supervised classifiers. This project implements kNN, Decision Tree/CART, and Logistic Regression. A stratified 80:20 train-test split is used so that the minority subscription class is represented in both partitions.

9.1 k-Nearest Neighbours

The kNN classifier predicts a customer from the labels of nearby training observations. Distance-based learning is sensitive to feature scale, so numerical variables are standardized. The implementation uses 15 neighbours and distance weighting as a reproducible baseline.

9.2 Decision Tree / CART

The Decision Tree recursively separates the observations using feature-based rules. A maximum depth and minimum leaf size are used to reduce excessive tree growth, while balanced class weights help the minority subscription class receive appropriate attention.

9.3 Logistic Regression

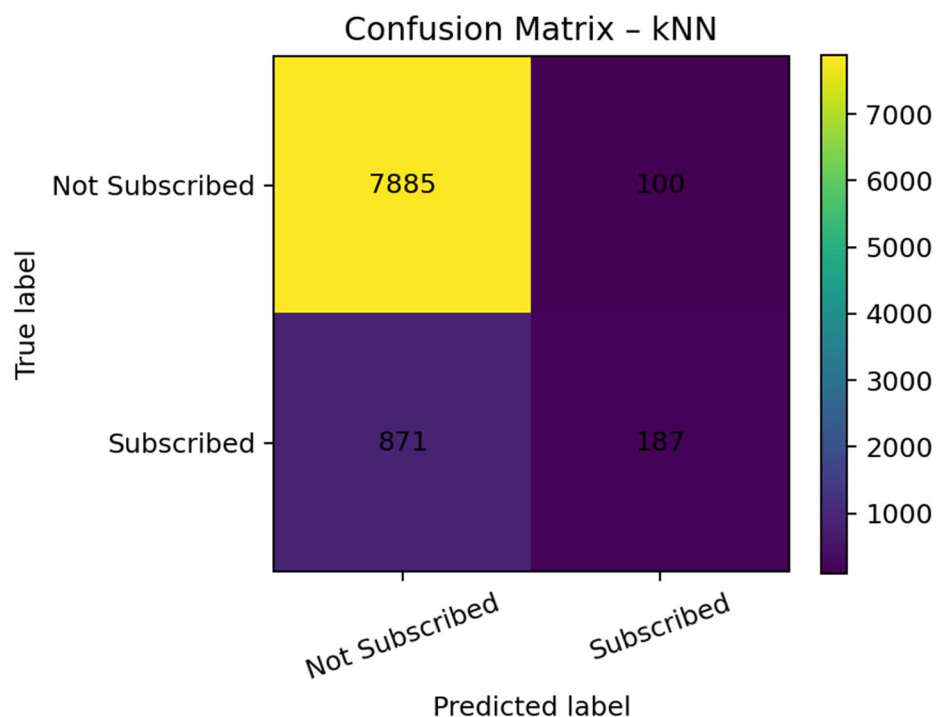
Logistic Regression estimates the probability of the positive class using a logistic link function. It provides a strong interpretable baseline for binary classification and uses balanced class weights in this implementation because the target classes are not evenly represented.

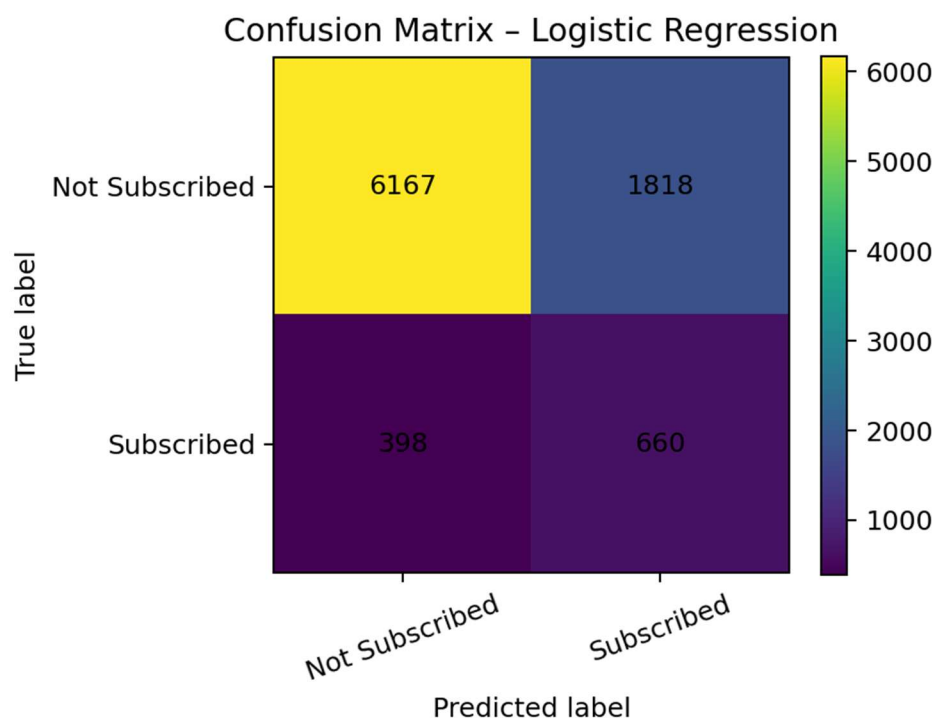
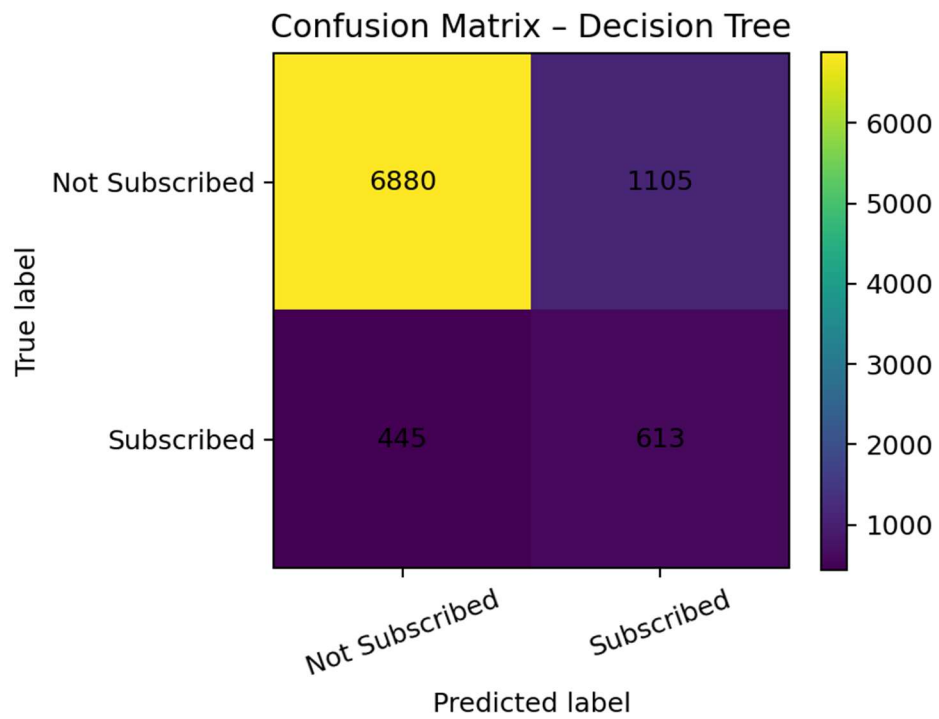
10. Model Evaluation and Selection

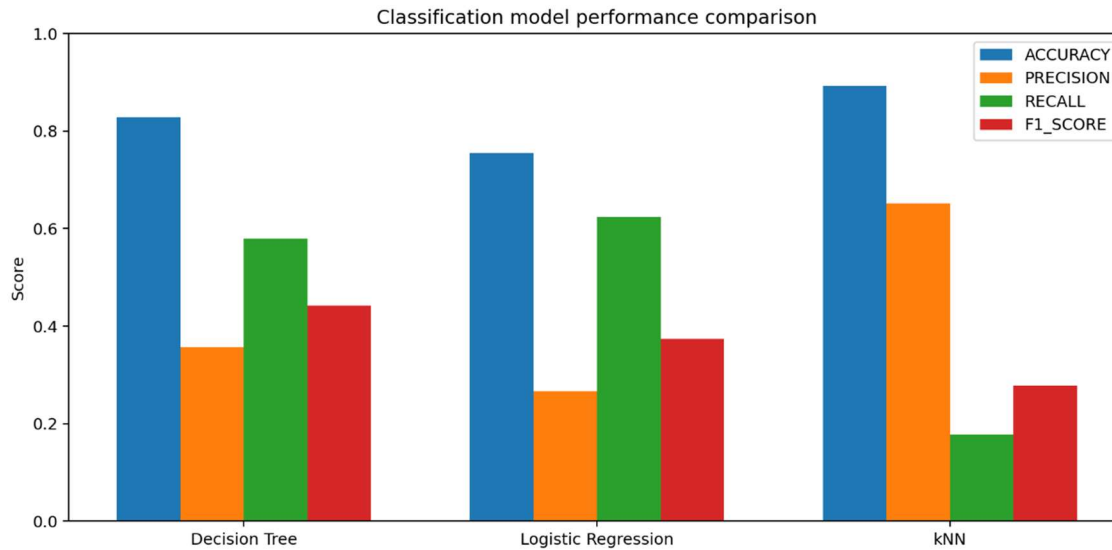
Accuracy is reported together with precision, recall, F1-score and ROC-AUC. The confusion matrix is used to inspect false positives and false negatives. Because the subscription class is smaller than the non-subscription class, a high accuracy score alone is not treated as sufficient evidence of a useful marketing model.

Model	Accuracy	Precision	Recall	F1_Score	Roc_Auc
Decision Tree	0.8285967046334182	0.35681024447031434	0.5793950850661626	0.44164265129682995	0.7792303148744161
Logistic Regression	0.75494857901139	0.26634382566585957	0.6238185255198487	0.3733031674208145	0.7722285286803114
kNN	0.8926241291606768	0.6515679442508711	0.1767485822306238	0.27806691449814125	0.7369037289909128

Best model: KNN. The selection should be justified using the actual scores in model_comparison.csv. In a marketing context, recall and F1 are especially useful when the objective is to find more potential subscribers without relying only on the dominant class.







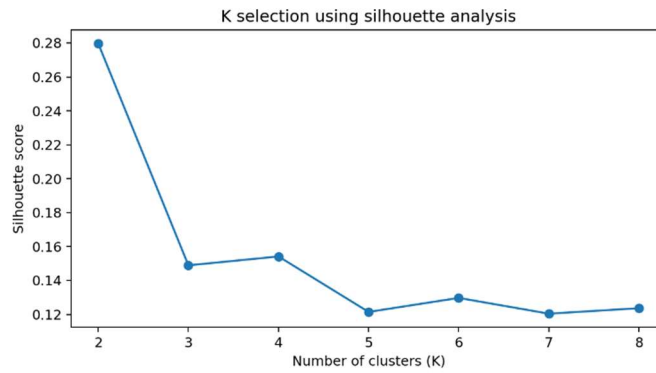
11. K-means Customer Segmentation

K-means is used to identify customer groups with similar demographic, financial and campaign characteristics. The clustering input contains age, balance, campaign contacts, previous contacts, pdays and selected categorical attributes such as job, education, housing, loan and previous campaign outcome.

11.1 Selecting K

Candidate values from K=2 through K=8 are evaluated using the silhouette score. The K with the highest score is selected as the primary segmentation solution. The exact score table and selected K are stored in outputs/kmeans_silhouette_scores.csv.

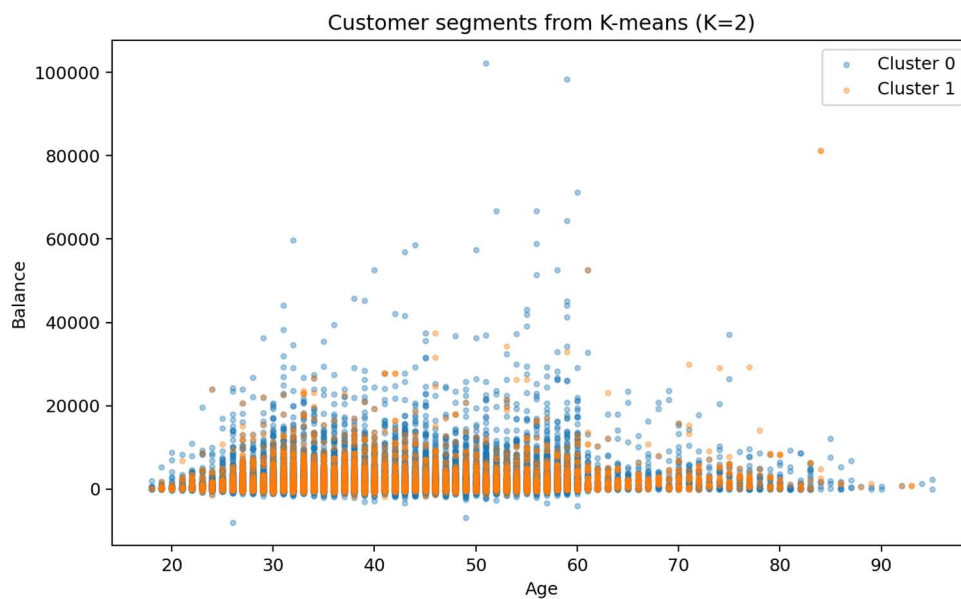
K	Silhouette score
2	0.27962225070580676
3	0.1489243107127213
4	0.15415453935608733
5	0.12146403601882143
6	0.12972668749219188
7	0.12043416797867038
8	0.12366192594521486



11.2 Cluster Profiles

For each cluster, the implementation reports customer count, mean age, mean and median balance, campaign-contact behaviour, previous contacts, and subscription rate. These values should be read from outputs/cluster_profiles.csv.

Cluster	Customers	Mean age	Mean balance	Campaign contacts	Subscription rate
Cluster 0	37518	40.960205767898074	1321.045604776374	417	2.9063116370808677
Cluster 1	7693	40.81918627323541	1563.329390354868	598	2.0690237878590927



12. Comparison of Segments and Predicted Subscription Behaviour

The final segmentation is not intended to replace the classifier. Instead, the two approaches answer different questions. Classification estimates whether an individual customer is likely to subscribe, whereas clustering describes groups of customers with similar profiles. The project compares the subscription rate of each cluster with the predicted behaviour of the selected classification model.

Segment interpretation	Evidence to identify	Recommended action
High-potential segment	High cluster subscription rate and high predicted-positive rate	Prioritize personalized offers and focused follow-up
High-value / financially strong segment	Higher balance with suitable subscription response	Use premium communication and product-specific messaging
Engagement-oriented segment	More previous contacts or stronger previous campaign outcomes	Use campaign history to personalize follow-up
Low-response segment	Low subscription rate and low predicted-positive rate	Reduce repeated contact and test alternative messaging
Young / lower-balance segment	Lower age and lower balance profile	Use simple, value-oriented digital communication

The final labels must be based on the actual cluster profiles rather than assumed characteristics. The table above gives the interpretation framework; the executed output determines which cluster number belongs to each business segment.

13. Data-driven Marketing Recommendations

Rank customers by predicted subscription probability and concentrate campaign resources on high-probability cases.

Use cluster profiles to vary message content instead of sending one generic offer to all customers.

Use previous campaign outcomes to distinguish customers with stronger historical engagement from low-response customers.

Avoid excessive repeated contacts for groups that show low response and high campaign contact counts.

Review model performance using recall and F1 before operational deployment because missed potential subscribers can reduce campaign effectiveness.

Periodically retrain the system as customer behaviour and campaign conditions change.

14. SDG Mapping

The problem statement on page 1 of the uploaded assignment maps the project to SDG 8, SDG 9 and SDG 12.

SDG 8 – Decent Work and Economic Growth: better targeted banking campaigns can support more efficient economic activity and customer-product matching.

SDG 9 – Industry, Innovation and Infrastructure: the project applies data science and machine learning to improve a real-world financial service process.

SDG 12 – Responsible Consumption and Production: targeted communication can reduce unnecessary repeated marketing contacts and use campaign resources more efficiently.

The rubric on page 5 contains wording referring to SDG 7/8/12. The report follows the explicit problem-statement mapping of SDG 8/9/12 and records the rubric wording separately rather than changing the official project mapping.

15. Additional Rubric Alignment

15.1 Neural Network Concepts and Learning Analysis

A perceptron is a basic linear unit that computes a weighted sum followed by an activation function. A multilayer perceptron extends this idea through hidden layers, allowing nonlinear decision boundaries. Backpropagation computes gradients of the loss with respect to network parameters and updates the weights using an optimization procedure. Neural networks are not required as one of the three classifiers in the technical requirements, so they are discussed conceptually rather than substituted for kNN, Decision Tree, or Logistic Regression.

15.2 Genetic Algorithms and Hypothesis-Space Search

A genetic algorithm represents candidate solutions as a population. Selection favours stronger candidates, crossover combines parts of candidate solutions, and mutation introduces variation. In model development, a genetic algorithm can be used to search a hypothesis or hyperparameter space when exhaustive search is expensive. The core implementation for this assignment uses deterministic scikit-learn classifiers and K-means, so a genetic algorithm is included here as a conceptual rubric-alignment discussion rather than an unnecessary fourth model.

15.3 Pandas and Menu-driven Functionality

The program demonstrates DataFrame inspection, filtering, sorting, grouping, aggregation and CSV export. An interactive menu provides dataset summary, subscription distribution, model comparison, cluster profiles, correlation output and a prediction workflow note. This addresses the implementation and reliability emphasis in the rubric on page 5.

16. Implementation Details

The implementation is contained in `bank_marketing_solution.py`. The program is modular: loading, preprocessing, descriptive analysis, inference, EDA, model training, clustering, summary generation and menu-driven functionality are implemented as separate functions. Random state 42 is used where reproducibility is relevant.

Required Python libraries: Pandas, NumPy, SciPy, scikit-learn and Matplotlib. The optional `ucimlrepo` package supports programmatic retrieval from UCI. The implementation does not depend on Seaborn so that the requested Matplotlib visualization requirement is directly satisfied.

17. Reflection: Design Decisions, Challenges and Learning

A major design decision was to keep duration for historical analysis but exclude it from the pre-campaign prediction model. This separates an analytically useful variable from a variable that can create leakage. Another decision was to report several classification metrics rather than treating accuracy as the only measure, because the target class is imbalanced. For clustering, silhouette analysis was selected because it gives a quantitative basis for choosing K instead of selecting the number of segments arbitrarily.

The main implementation challenges are expected to be categorical encoding, class imbalance, choosing a useful clustering representation, interpreting outliers without deleting valid banking observations, and connecting technical outputs to marketing decisions. The project provides practical experience in converting a raw dataset into a reproducible data science pipeline and in interpreting model outputs in a business context.

18. Conclusion

The proposed system provides a complete workflow for bank customer subscription prediction and customer segmentation. It satisfies the core assignment requirements by combining preprocessing, EDA, descriptive statistics, statistical inference, three supervised classifiers, evaluation, K-means clustering, visualization, and business recommendations. The final numerical conclusions must be populated from the executed outputs so that every reported metric and cluster description corresponds to the actual dataset used.

19. References

- Moro, S., Rita, P., & Cortez, P. (2014). Bank Marketing [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C5K306.
- Moro, S., Cortez, P., & Rita, P. (2014). A data-driven approach to predict the success of bank telemarketing. *Decision Support Systems*, 62, 22–31.
- Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Harris, C. R., et al. (2020). Array programming with NumPy. *Nature*, 585, 357–362.
- McKinney, W. (2022). *Python for Data Analysis* (3rd ed.). O'Reilly Media.
- Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. *Computing in Science & Engineering*, 9(3), 90–95.

20. Individual Contribution Write-up

Complete this one-page section separately for each team member. The assignment requires an individual contribution write-up. Replace the placeholders with the student's actual work; do not claim work that was not performed.

Individual reflection: Describe the part of the implementation personally completed, the difficulty encountered, the debugging/testing performed, and the main data science concept learned.

Appendix A – Output Files

preprocessing_summary.csv

descriptive_statistics.csv

covariance_matrix.csv

correlation_matrix.csv

selected_covariance_correlation.csv

confidence_interval.csv

job_group_summary_sorted.csv

model_comparison.csv

kmeans_silhouette_scores.csv

cluster_profiles.csv

customer_segments.csv

execution_summary.txt

PNG figures in outputs/figures/

Appendix B – Originality and Plagiarism Control

This report has been drafted specifically for the uploaded DSA0402 assignment rather than copied from a pre-existing project report. External facts and the dataset are attributed to their sources. Code is written as an original implementation for this project. A similarity score below 20% cannot be guaranteed because institutional plagiarism tools use different databases, settings, and thresholds. Before submission, the student should run the completed document through the institution's approved plagiarism checker and revise any highlighted overlap that is not a necessary technical term or citation.